Statement of Purpose

Mahir Desai

The idea to pursue AI/ ML came about when I began my studies for the degree of B.Tech in Computer Science at Manipal University Jaipur. Through my journey with the field, I came to know that the construction of efficient ML systems demands sound knowledge of mathematics, statistics, and data science. To enhance my foundations in the above disciplines, I concurrently pursued the degree of B.S. in Data Science and Applications from IIT Madras.

My work in classical ML included various projects on predictive models, time-series forecasts, and fraud detection. In the Fraud Detection under Concept Drift project, I constructed machine learning pipelines for financial transaction datasets using temporal validation, feature engineering, and interpretability methods.

As my interests evolved, I moved onto deep learning and biomedical AI. For my research, I worked as an Undergraduate Student Researcher at Manipal University Jaipur on designing a Parkinson's disease classification system based on EEGNet architectures using subject-wise validation method. The research has been published in a paper at ICICC 2026 (Springer) and I am the first author of the paper.

During the past year, my interests turned toward Agentic AI and intelligent systems. As such, I have created ResearchIQ, an AI-powered research retrieval engine which incorporates Semantic Scholar, OpenAlex, and CrossRef with semantic search and retrieval pipelines. I also designed MediWagon, an AI healthcare assistance system using LangGraph-based agent workflow, as well as Slide2Learn which creates summaries and quizzes for educational content using LLM-powered workflows.

To extend my exploration of NLP, I designed Multilingual Review Intelligence, a system for analyzing sentiment of English and Hindi reviews using traditional machine learning models as well as transformer models. I gained experience with Computer Vision by designing Aegis, a privacy detection system using YOLOv8, OCR, face recognition, and risk assessments for identification and obfuscation of private details in live video feeds.

Presently, I am a Summer Research Intern at IIT Kharagpur working on designing an Edge AI system for classification of ECG recordings for ventricular tachycardia and fibrillation detection. My work involves aspects of biomedical signal processing, deep learning, TinyML, android development, and real-time deployment of a complete AI system.

Whereas I have already developed ML projects and done research, I would like to explore more theoretical aspects of machine learning, such as optimization, theory, representation learning, scaling ML applications and real-world deployment problems. Amazon ML Summer School is an amazing chance to learn from the scientists who have implemented advanced ML systems on an industrial level.

My future career plans include working on healthcare technology, as well as exploring other fields that require analytical skills and data-based decision making. My interest in research and my ability to engineer solutions make me a perfect candidate for this summer school.